UCSF Radiology

RISE

Radiology Initiative for Scholarly Engagement

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Department of Radiology & Biomedical Imaging,

"CT Assessment of Hepatic Vasculature for ChemoFilter Treatment Planning and Further Device Development"

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Collaborators & Lab Members: Colin Yee, Teri Moore, Bridget Kilbride

Introduction

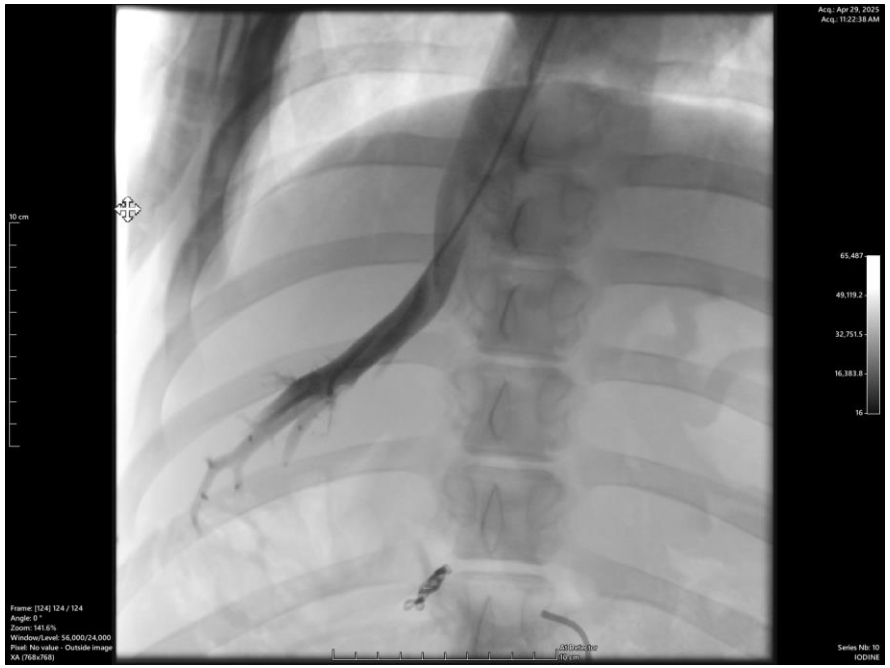
Hepatocellular Carcinoma

80%-90% of primary liver cancer cases

4th most common cause of cancer death

Hepatic Intra-Arterial Chemotherapy (IAC)

image-guided, local administration of chemotherapeutic agents to arteries feeding the tumor

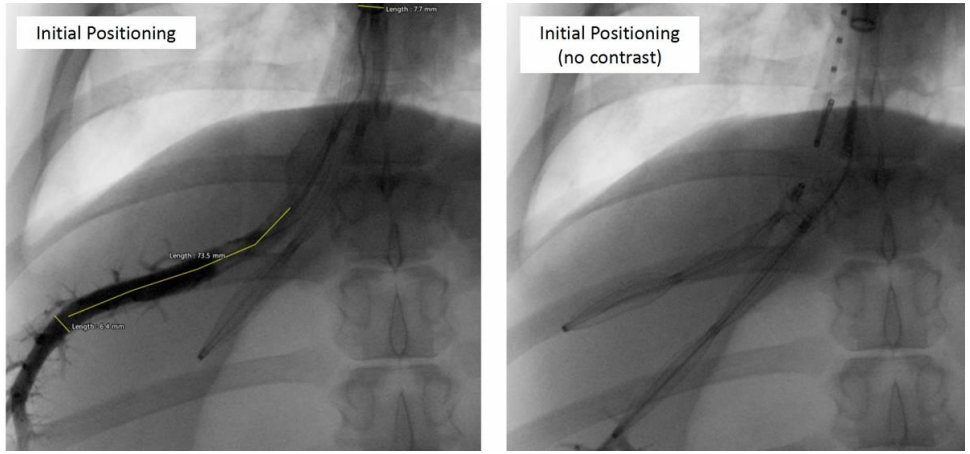




Introduction

Doxorubicin (DOX)

- Effective chemotherapeutic agent
- Side effects
  - Off-target accumulation in heart and kidneys
  - Cardiomyopathy risk
- Dose limitations to mitigate side effects
  - Reduction in antitumor effect

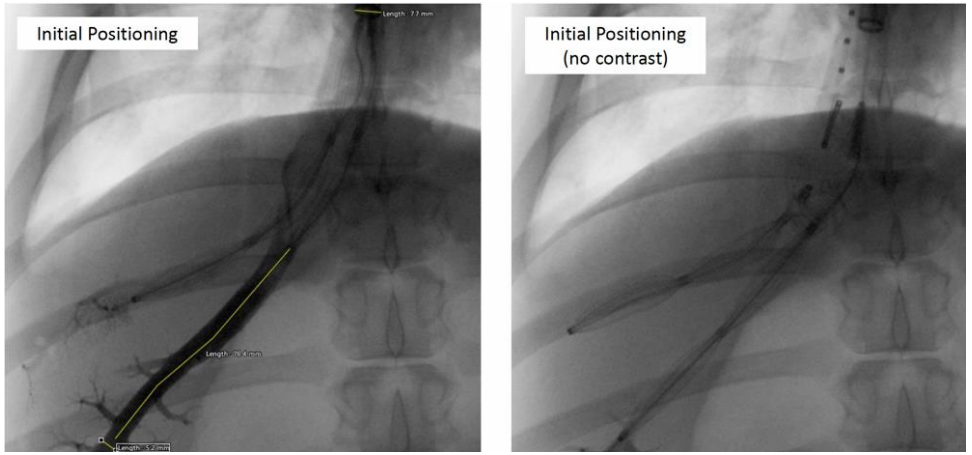


ChemoFilter

- Ionic device deployed in and adjacent to hepatic venous outflow tract to reduce off-target DOX exposure

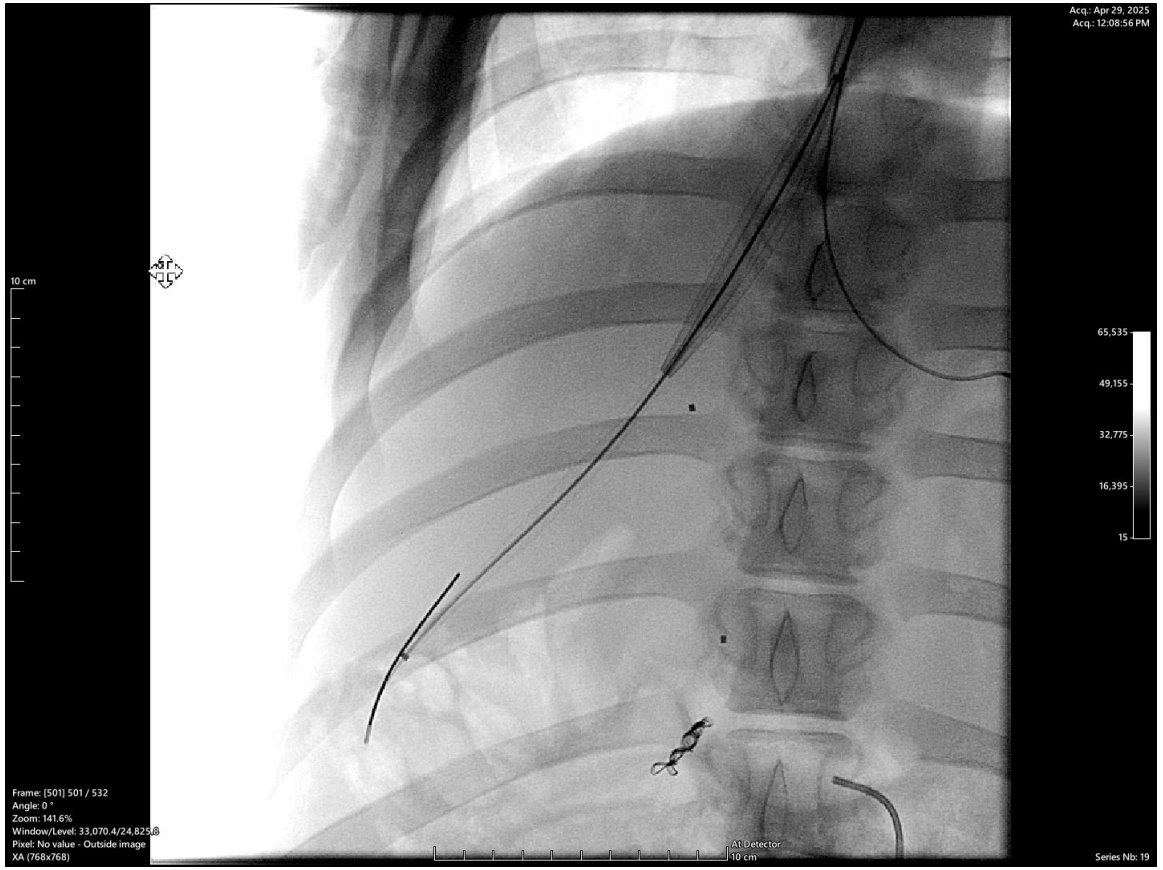
Animal Study Results

- 31% decrease in peak DOX concentration
- Increase in time to maximum concentration
- Reduced DOX accumulation in heart and kidneys

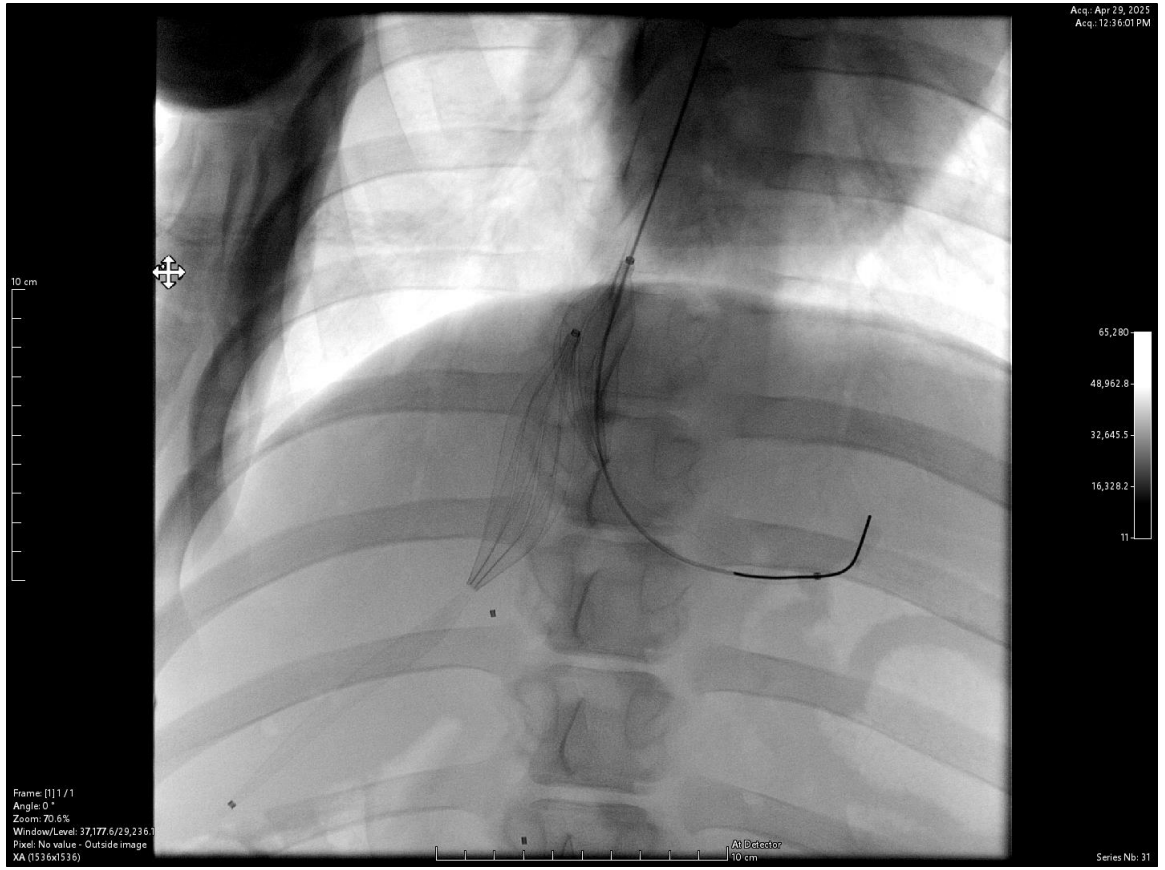


Deployment of ChemoFilter

Elongated



Expanded



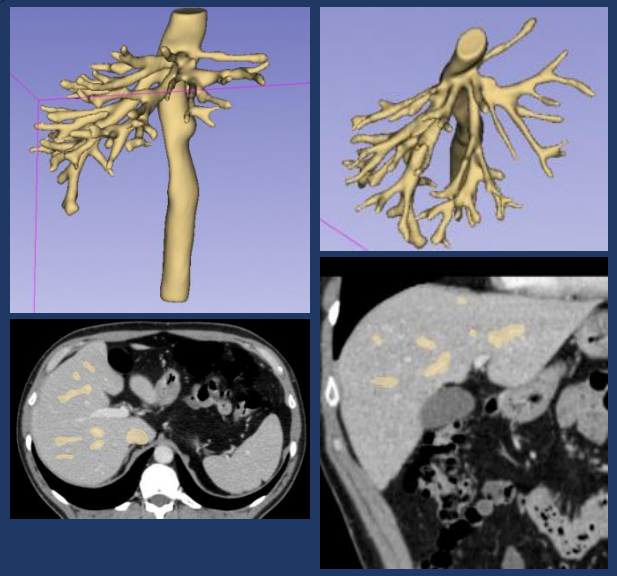
Problem Statement

Using patients' abdominal CT scans to create 3D models of hepatic vasculature to improve treatment planning

- Improves deployment position of ChemoFilter devices (selection of accessible hepatic veins and location for optimal filtration)
- Reduces treatment time (less operating time spent on visualization and navigation)

Methods

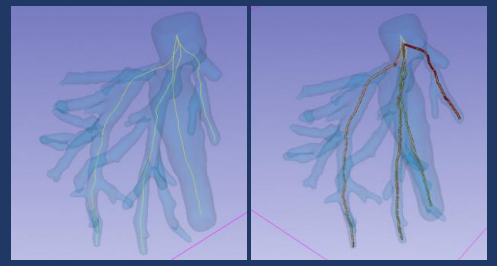
Segmenting Hepatic Vasculature



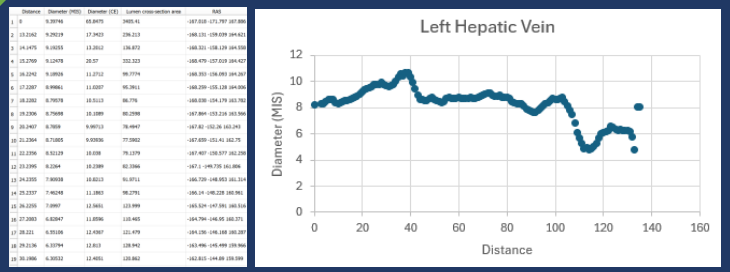
- 3D Slicer Tools for Segmentation
  - Cropping CT Volume, Filtering to Denoise, Thresholding, and Smoothing
- Hepatic Vasculature
  - IVC, left hepatic vein, middle hepatic vein, and right hepatic vein

Extracting Features

- Centerline Extraction



- Cross-Sectional Analysis
  - Diameter & Lumen Cross-Section Area Per Distance from Root

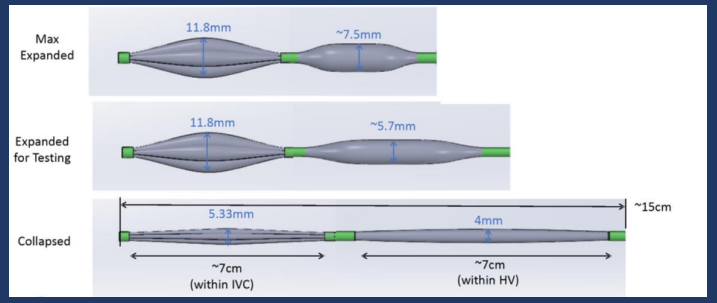


- Measuring Angle of Entry



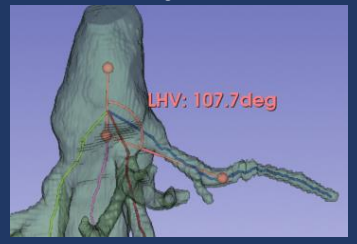
Analysis

- Analyzing the Diameter-Distance Plot to Determine if the Device Fits

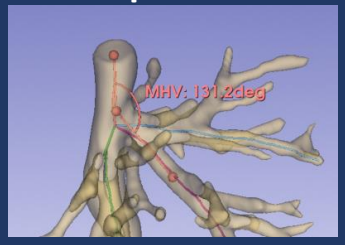


- Suboptimal Angle of Entry: 90°- 110°
  - determined empirically due to device mechanics (rigid) & lab observations in previous animal studies

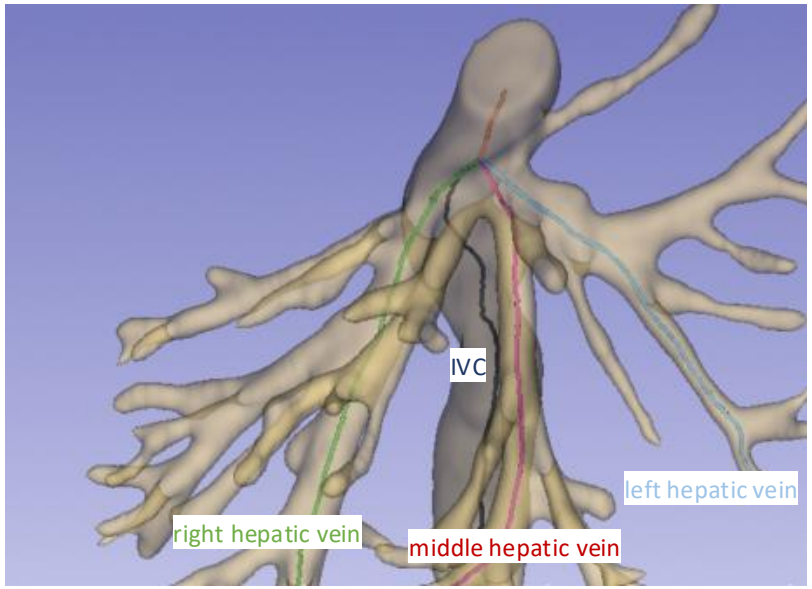
Suboptimal

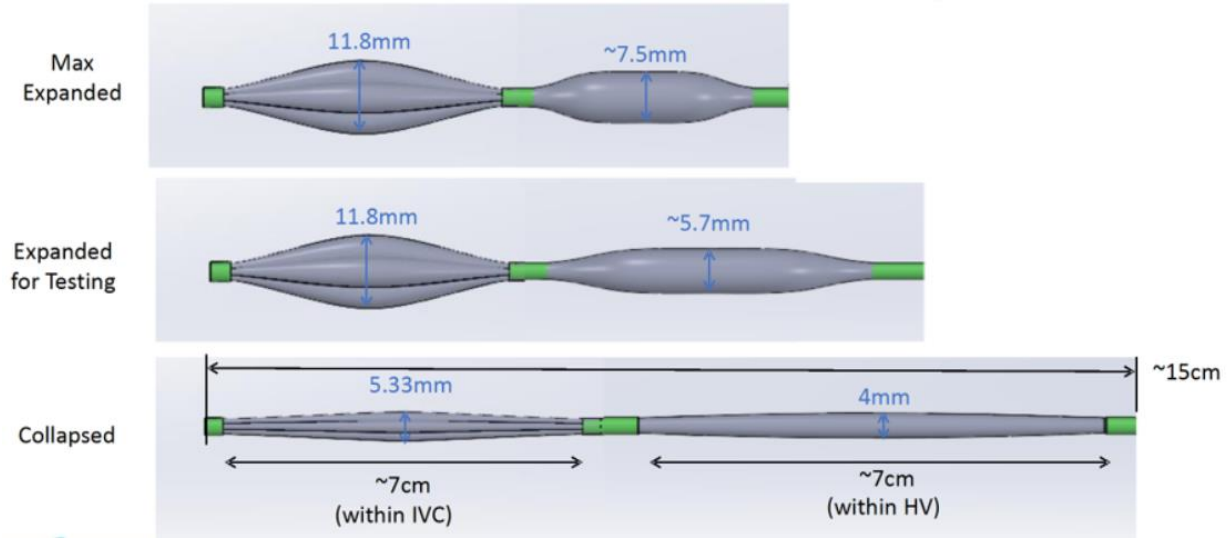


Optimal

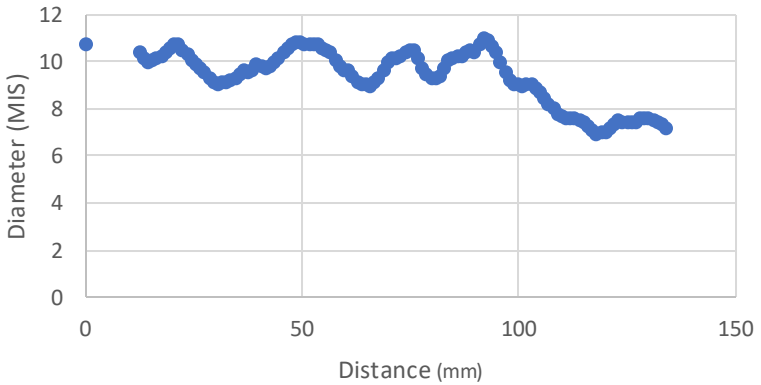


Results: Case Study on Liver Sample #11





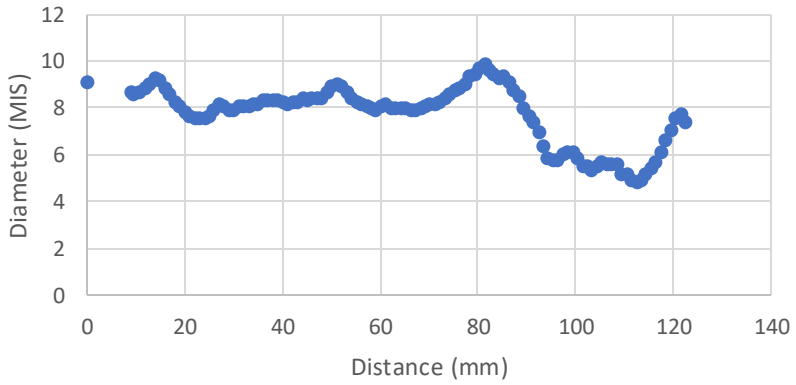
right hepatic vein



angle of entry: 146.6°

Optimal Dimensions

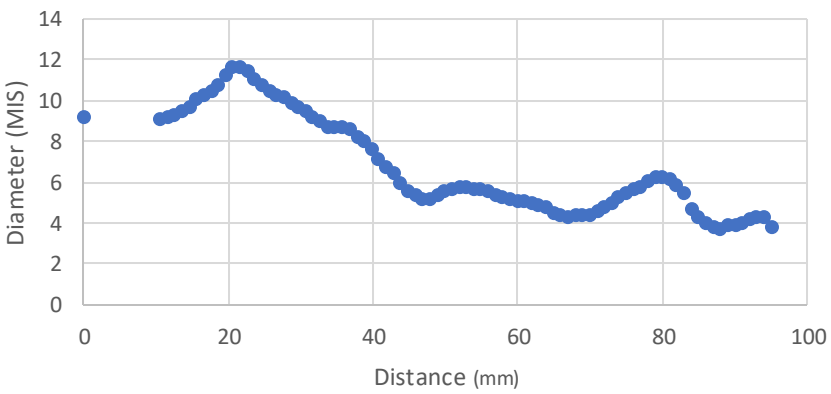
middle hepatic vein



angle of entry: 131.2°

Optimal Dimensions

left hepatic vein



angle of entry: 94.8°

Suboptimal Angle of Entry & Dimensions

Discussion

Impact of Work

- We created a protocol for analyzing abdominal CT data to assess patients' hepatic vasculature for ChemoFilter treatment.
  - 3D printing models for pre-treatment planning
  - Determining accessibility of hepatic veins to reduce navigation time in OR
- We created a database of hepatic anatomy for future device development.
  - Building future prototypes that better improve accessibility for currently suboptimal anatomies, thus increasing the number of patients that can benefit from ChemoFilter

Limitations

- Small number of patient CT scans analyzed so far
- Lower resolution of scans used; with higher resolution scans, we can improve the accuracy of our models
- Need to improve metrics used to evaluate hepatic vein accessibility